Machine Learning Models for Predicting Cardiac Mortality in Chronic Heart Failure Patients

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***Abstract*—Sudden cardiac death (SCD) and pump failure death (PFD) are common risks for people with chronic heart failure (CHF). In this paper, we evaluate machine learning approaches for predicting cardiac mortality, including SCD and PFD, in patients with CHF using a tabular dataset from the publicly available MUSIC (MUerte Subita en Insuficiencia Cardiaca) database. The tabular data includes clinical and demographic variables, imaging and laboratory measurements, and markers of cardiac electrical activity derived from two different types of electrocardiogram tests. Our study investigates the best-performing model configurations for five classifiers (Random Forest, Naïve Bayes, Logistic Regression, Support Vector Machine, and K-Nearest Neighbors) under different hyperparameter settings and data processing methods. Across the classifiers, we vary the feature selection technique, number of selected features, and random undersampling ratios. We perform ANOVA and Tukey's HSD test to evaluate significant differences for each factor. The best-performing models used Random Forest with either Information Gain or Chi-Squared feature selection, 10, 50, or 100 features, and a perfectly balanced class ratio, which resulted in AUC and AUPRC scores ranging from 0.73 to 0.78. Overall, our study highlights the potential of applying machine learning methods to the MUSIC database for future research on predicting SCD and PFD in patients with CHF.**

***Index Terms*—Cardiac health, Heart failure, Sudden cardiac death, Machine learning, Feature selection, Random Undersampling**

I. Introduction

Chronic heart failure (CHF) is a long-term condition which affects around 6.7 million people in the United States alone and is cited in over 14% of death certificates in 2023 [1]. It is a condition in which not enough oxygenated blood is being pumped, which could lead to a sudden cardiac death (SCD) or pump failure death (PFD). SCD is defined as a death caused directly by a heart-related issue, such as arrhythmia, usually within 60 minutes from the onset of witnessed symptoms [2]. This event is responsible for over 180,000 deaths in the United States per year [2]. PFD is defined as a death due to the heart progressively becoming unable to pump enough blood to stay alive and is a common outcome of end-stage heart failure [3], [4]. Researchers can leverage machine learning (ML) models to identify risk factors and improve prediction of cardiac mortality [5]. A recent systematic review highlights the potential of ML models for SCD prediction using multidimensional data over traditional methods [6].

This study aims to determine an optimal model configuration for predicting the likelihood of a cardiac death (SCD or PFD) in CHF patients. We used the MUSIC (MUerte Subita en Insuficiencia Cardiaca) database, which was recently made publicly available and is split into multiple modalities, including tabular and signal data for up to 992 patients with symptomatic CHF [7]. Outcomes at the end of the follow-up period include SCD, PFD, non-cardiac death, and survival. The tabular dataset, which we focus on in this work, consists of clinical and demographic variables, radiographic and laboratory measurements, medications taken at time of enrollment, and markers of cardiac electrical activity derived from a 20-min, high-resolution electrocardiogram (ECG) and 24-hour ECG recorded using a Holter monitor. Overall, the MUSIC dataset is a high-quality resource for predicting mortality, providing a detailed profile of the study patients through its ECG-derived variables, clinical parameters, and ECG signal recordings.

In this paper, we analyzed the tabular dataset to evaluate how different ML configurations affect prediction of cardiac mortality. The data was first cleaned and preprocessed by removing irrelevant features, imputing missing data, and encoding categorical features. Then, we ran five classifiers—Random Forest, Naïve Bayes, Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)—in configurations that varied by feature count, feature selection technique, and level of random undersampling (RUS). For Random Forest, we also explored a range of values for maximum tree depth and number of trees. The results from all models were then examined to determine the best-performing configuration for the dataset. To the best of our knowledge, no prior work has systematically evaluated multiple ML models for the classification of cardiac mortality using the tabular dataset from the MUSIC study. Our work presents the first comprehensive analysis of classifier performance across varying feature selection methods, feature counts, RUS ratios, and hyperparameter configurations with the MUSIC data. We also briefly summarize the most informative variables under the top-performing feature selection methods.

The remainder of the paper is organized as follows: Section II discusses a sample of related works that have used the MUSIC dataset; Section III discusses the data processing and modeling; Section IV presents our findings and the best model configurations as well as an overview of the top features; and Section V concludes our study and presents directions for future work.

II. Related Work

A previous survey shows that the application of machine learning for SCD prediction is underexplored, but that the studies it reviewed suggest advanced ML models may provide more insights than traditional tabular data analysis [6]. The MUSIC dataset has many advantages, both in its variety of patient data and its potential utilization. The database became publicly available in 2024 and has been used in a limited number of studies. The large number of available features and multiple modalities allows for broader research on attribute associations to cardiac death, as well as a potential for increased performance in predictive modeling. A list of papers that use the dataset is available on the database's PhysioNet page [8]. We conducted a thorough survey of works that applied the MUSIC database. No previous work has performed a ML classification task using the MUSIC tabular data for direct comparison. Instead, the majority of studies focus on risk stratification using biomarkers directly derived from the ECG recordings. Only one paper focused on risk scores based on the tabular data alone [3], while another work compared the performance of models built using different groups of variables within the MUSIC dataset [9]. While these do not involve machine learning, the two studies show examples of how to use the tabular data from the MUSIC database for statistical analyses and risk models. Due to space limitations, we suggest consulting the related works for definitions of medical terminology and acronyms used throughout this paper.

Vazquez et al. [3] used multivariable Cox proportional hazards models to produce a risk score of different types of mortality rates. The authors only processed the tabular data available in the MUSIC database as input to Cox models, with the outcomes being mortality from any cause, cardiac mortality, SCD, and PFD. The primary target of these experiments was cardiac mortality, while SCD and PFD were secondary targets. They then developed a MUSIC Risk score by multiplying each variable by its β-coefficient for each of the models and then summing the products. Each risk score was then compared with its respective probability of death, which was calculated by the Cox models. This was done for each outcome, and distinct risk groups were found. With their four final models, only subsets of 5 to 9 features were used and taken from the following group: prior atherosclerotic vascular event, left atrial size $>$ 26 mm/m2, ejection fraction $\leq$ 35%, atrial fibrillation, left bundle branch block or intraventricular delay, non-sustained ventricular tachycardia and frequent ventricular premature beats, estimated glomerular filtration rate $<$ 60 mL/min/1.73m2, hyponatremia $\leq$ 138 mEq/L, NT-proBNP $>$ 1 ng/L, and troponin-positive. The correlation between the estimated survival rate and the actual survival rate was very strong at 0.99. Calculating the c-indices for each type of mortality, they found values ranging from 0.74 to 0.8, which are also decently strong results. They categorized those with risk scores of above 20 as high-risk patients and those with below 20 as low-risk patients. Overall, the authors were able to show the usefulness of their proposed method in being a non-invasive method for predicting mortality rates for patients with CHF. Although the study provides risk scores for SCD and PFD, the authors recommend first evaluating the overall cardiac mortality risk score and using the SCD and PFD scores only for patients identified as high-risk.

The authors selected features associated with increased cardiac mortality, and additional variables within the dataset were not explored. As a result, there may be other variables available in the MUSIC dataset that have unknown predictive relationships with the outcomes. Our study addresses this gap by systematically evaluating varying numbers of features, including the full set, using multiple feature selection techniques. This approach enables a more objective assessment of which features and what number of features provide the greatest predictive value for the classification task.

Ramírez et al. [9] predicted the risk of SCD or PFD using multivariable Cox models given clinical and ECG variables in the dataset. For their first model, they chose a subset of clinical features based on known relationships to SCD or PFD, which include patient age, their gender, their New York Heart Association (NYHA) class, diabetes status, and beta-blocker use. They then extracted additional ECG features from the existing ECG tabular data, such as repolarization restitution and spatio-temporal dispersion of repolarization. They combined these with selected ECG features, including the RR range and QRS $>$ 120 ms, for their second model. Lastly, they implemented a third model that uses both clinical and ECG features. Models using both clinical and ECG features achieved the best performance, followed by ECG-only models, while clinical-only models performed the worst. For SCD, using both feature subsets produced a significant improvement in the performance compared to using either subset alone. For PFD, using both data yielded slightly better but similar results compared to using the ECG-only data.

The authors considered only a small subset of the clinical and ECG features available in the MUSIC dataset, which restricts the number of potential predictive relationships. It is possible that by using a larger group of pre-existing features, the models may derive new associations or have an increased performance when only using clinical data. In contrast, we consider a majority of the independent features available in the dataset, excluding only variables that are directly related to study logistics or provide redundant information.

The novelty of our work lies in our in-depth comparison of various classifier models and configurations. We systematically evaluate a wide range of preprocessing options and model hyperparameters, allowing us to analyze how different classifiers, feature selection methods, feature counts, and RUS ratios influence predictive performance. Additionally, while prior works have focused on traditional statistical analyses of risk models and predictors, we investigate the application of ML models with the MUSIC database, using its extensive tabular dataset. This study provides a broad assessment of how ML approaches can be applied to predict cardiac mortality outcomes in patients with CHF.

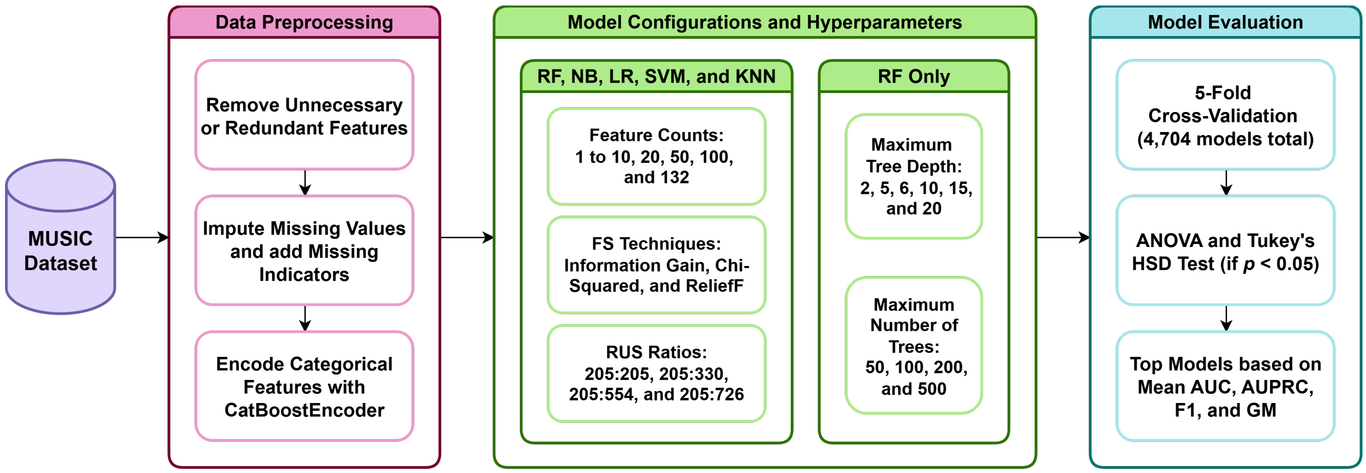

Fig. 1. Modeling pipeline for the MUSIC dataset, including preprocessing, model configurations, and evaluation. FS represents feature selection.

III. Methodology

The MUSIC database contains both tabular and signal data for patients with CHF [7]. All patients were followed at six-month intervals for a median duration of 44 months, providing a robust dataset for long-term mortality prediction. The tabular dataset contains 992 patient samples with 103 numerical and categorical features. The database also contains high-resolution ECG signal data for 691 patients and 24-hour Holter ECG signal data for 936 patients. We do not utilize the signal data, as it is outside the scope of this study and cannot be explored due to space constraints.

According to the "Cause of death" target attribute in the tabular data, 726 patients were labeled as survivors, 61 had a non-cardiac-related death, 94 died due to SCD, and 111 died due to PFD. This shows that the dataset suffers from class imbalance for the primary outcomes of SCD and PFD.

All the experiments in this paper were done using Python 3. The libraries pertinent to ML experimentation that we use are pandas, numpy, matplotlib, scikit-learn, and scipy.

A. MUSIC Data Modeling

Prior to modeling, the tabular data was preprocessed by removing unnecessary features, imputing missing values, and encoding categorical features. First, the unique identifier and target attribute were removed from the dataset. Then, any study or logistic attributes were omitted, including the follow-up period length features, the exit of the study, the availability indicators for the high-resolution and Holter ECG data, and the Holter onset time. Lastly, any redundant features were removed such as data that is covered by other attributes including the body mass index (BMI), QRS $>$ 120 ms, and the RR range. All instances with a non-cardiac death outcome were removed from the dataset to focus on predictors for SCD and PFD. The target variable "Cause of death" was redefined into a binary feature, where the value 1 represents that the patient either died of SCD or PFD, and 0 represents the patient survival by the end of the follow-up period. We chose to combine the SCD and PFD into a single cardiac mortality outcome to center the analysis on overall cardiac death risk among patients with CHF. Because both outcomes represent major cardiac causes of death in this population, combining them provides a clinically meaningful endpoint while increasing the number of positive cases. All missing numerical features were imputed using mean values with missing indicators and all missing categorical features were given the label "Unknown" and encoded using a CatBoost-based encoder [10], [11]. The result is a dataset with 931 instances and 132 features.

After initial preprocessing, we apply many different model configurations to determine which hyperparameters achieved the best performance. In particular, we selected a wide range of values, allowing us to explore trade-offs between model simplicity and performance. First, we examine feature selection, the process of removing unhelpful or redundant features, using three techniques: Chi-Squared [12], ReliefF [13], and Information Gain [14]. Each technique scores the features, from which we select different numbers of top features: 1 to 10, 20, 50, 100, and the full set of 132 features. Different levels of RUS were also compared by pairing the full set of 205 positive cases with varying numbers of randomly chosen negative cases: 726, 554, 330, and 205. We chose five commonly used ML classifiers: Random Forest, Naïve Bayes, Logistic Regression, SVM, and KNN. Prior to the main experiment, we conducted a preliminary analysis to select the K value for KNN by evaluating potential values (3, 5, 7, and 9) using stratified 5-fold cross-validation. The K values of 7 and 9 achieved the highest mean performances and were statistically different from the other two K values. We chose to use $K = 7$ for all KNN models in the main experiment, since it is less computationally expensive than $K = 9$. Random Forest was run using maximum tree depth values of 2, 5, 6, 10, 15, and 20, and a set number of trees of 50, 100, 200, and 500. In total, we investigate 4,704 model types and configurations, which are each evaluated under 5-fold cross-validation. The overall modeling pipeline is shown in Figure 1.

B. Performance Evaluation

The performance metrics used were false negative rate (FNR), false positive rate (FPR), area under the receiver operating characteristic (ROC) curve (AUC), area under the precision-recall curve (AUPRC), F-measure, and G-mean. We selected these metrics to include both threshold-dependent and threshold-independent measures. FNR, FPR, F-measure, and G-mean are calculated at a fixed classification threshold, whereas AUC and AUPRC are threshold-independent metrics that summarize model performance across all possible thresholds [15]. For all of our experiments, we use the default classification threshold of 0.5.

When choosing the optimal models, only the AUC score, AUPRC score, F-measure, and G-mean metrics were considered. The AUC score represents the area under the ROC curve that plots the true positive rate (TPR) against the FPR across classification thresholds. The AUPRC score represents the area under the curve that plots the precision versus the recall of a model across classification thresholds. Precision is defined as the proportion of true positive predictions out of all positively predicted cases. Recall is measured as the proportion of true positive predictions out of all actual positive cases and is equivalent to TPR. The F-measure is calculated through the following formula:

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \tag{1}$$

Meanwhile, the G-mean refers to the geometric mean of the true positive rate and true negative rate (TNR):

$$GM = \sqrt{TPR \cdot TNR} \tag{2}$$

Analysis of variance (ANOVA) and Tukey's Honestly Significant Difference (HSD) tests were performed to determine which classifiers, feature counts, feature selection methods, and RUS ratios differed significantly in performance. Separate groupings were reported based on each of four metrics: AUC, AUPRC, F-measure, and G-mean.

IV. Results

A. HSD Analysis

Given the resulting 4,704 model types and configurations, a full report of all performances would be impractical. Therefore, our analysis focuses on the statistically significant effects of experimental factors across models. ANOVA showed statistically significant differences in five out of the six factors (classifier, feature count, feature selection technique, RUS ratio, and maximum tree depth) at $p < 0.05$. The number of trees for Random Forest proved to be the only non-significant ANOVA result. Tukey's HSD tests were subsequently applied to measure the significance of the independent configurations and order them in groups based on the significance of their performances. For example, if Value 1 is placed in group "a" and Value 2 is placed in group "b", that means that Value 1 is significantly better than Value 2. Configurations sharing one or more letters are not statistically significantly different at the selected significance level. In Tables I through V, all instances of group "a" are in bold.

The first factor analyzed was the choice of classifier. Table I shows the group rankings for each classifier used in the experiments. The top-performing classifier was consistently Random Forest for all performance metrics, while Naïve Bayes showed comparable performance for AUPRC, F-measure, and G-mean. Logistic Regression appeared in the top-performing group only according to AUPRC and F-measure, while KNN overlapped with the top group for F-measure and G-mean but performed significantly worse for AUC. On average, SVM performed significantly below most of the classifiers across the metrics. Overall, these results indicate that Random Forest provides the most consistently strong performance across metrics, followed by Naïve Bayes. KNN and Logistic Regression may achieve statistically comparable results for select measures.

TABLE I
TUKEY'S HSD GROUPINGS OF CLASSIFIERS BASED ON AUC, AUPRC, F-MEASURE, AND G-MEAN

Classifier	**AUC**	**AUPRC**	**F1**	**GM**
RF	**a**	**a**	**a**	**a**
NB	b	**a**	**a**	**a**
LR	c	**a**	**a**	b
SVM	d	b	b	c
KNN	d	b	**a**	**a**

The next factor was the number of features chosen by the feature selection technique, as shown in Table II. The feature counts consistently assigned to the top group across all metrics are 50 and 100. This means that in this context, it is best to choose either the 50 or 100 highest-ranked variables when applying feature selection techniques, since it reduces training time while maintaining a top performance. However, if relying on either F-measure and G-mean, a lower feature count of 4 to 20 may provide a strong mean performance that is not significantly different from that of a higher feature count.

TABLE II
TUKEY'S HSD GROUPINGS OF FEATURE COUNTS BASED ON AUC, AUPRC, F-MEASURE, AND G-MEAN

Feature Count	**AUC**	**AUPRC**	**F1**	**GM**
1	j	h	d	e
2	i	g	cd	de
3	h	f	bc	cd
4	g	ef	**ab**	**abc**
5	f	def	**ab**	**abc**
6	e	cde	**ab**	**abc**
7	d	bcd	**ab**	**abc**
8	cd	bc	**ab**	**ab**
9	c	b	**a**	**a**
10	c	b	**a**	**a**
20	b	**a**	**a**	**a**
50	**a**	**a**	**a**	**ab**
100	**a**	**a**	**ab**	**abc**
132	**a**	**a**	**ab**	bc

For feature selection techniques, Table III shows the groupings for each option. Information Gain is consistently in group "a" across the four metrics. Meanwhile, Chi-Squared has a statistically similar mean performance for AUPRC, F-measure, and G-mean. Lastly, ReliefF is always in the lowest-performing group for each metric. For this context, Information Gain or Chi-Squared is recommended.

TABLE III
TUKEY'S HSD GROUPINGS OF FEATURE SELECTION TECHNIQUES BASED ON AUC, AUPRC, F-MEASURE, AND G-MEAN

Feature Selection	**AUC**	**AUPRC**	**F1**	**GM**
Information Gain	**a**	**a**	**a**	**a**
Chi-Squared	b	**a**	**a**	**a**
ReliefF	c	b	b	b

Next, the RUS ratios were analyzed for statistical significance, with the results presented in Table IV. Across all performance metrics, the top-performing model always used a ratio of 205:205. These results suggest that, when predicting cardiac mortality using the MUSIC dataset, a balanced 1:1 ratio may produce the highest mean performance.

TABLE IV
TUKEY'S HSD GROUPINGS OF RUS RATIOS BASED ON AUC, AUPRC, F-MEASURE, AND G-MEAN

RUS Ratio	**AUC**	**AUPRC**	**F1**	**GM**
205:726	b	d	d	d
205:554	c	c	c	c
205:330	c	b	b	b
205:205	**a**	**a**	**a**	**a**

Lastly, Table V shows the HSD test results for maximum tree depth. Although this experimental factor is specific to the Random Forest classifier, we chose to include the HSD groupings due to a statistically significant ANOVA result. No single maximum depth placed in the top-performing group across all four metrics. These findings suggest that optimal maximum depth depends on the metric used, and we recommend evaluating a range of depth values and determining the optimal configuration based on performance and desired classification behavior.

TABLE V
TUKEY'S HSD GROUPINGS OF MAXIMUM TREE DEPTHS BASED ON AUC, AUPRC, F-MEASURE, AND G-MEAN

Max Depth	**AUC**	**AUPRC**	**F1**	**GM**
2	**a**	**a**	d	c
5	**a**	**ab**	c	b
6	**a**	**abc**	bc	b
10	b	bc	**ab**	**a**
15	b	c	**a**	**a**
20	b	c	**a**	**a**

B. Optimal Models

A subset of models was chosen using only the top-performing options. Models were run using either Random Forest or Naïve Bayes, consisting of 4 to 100 features selected by Information Gain or Chi-Squared. We did not perform Tukey's HSD test for the number of trees, since ANOVA did not indicate significant differences in mean performance for this factor. Because the HSD results did not show a maximum tree depth that was consistently superior across the metrics, additional Random Forest configurations were tested by varying the maximum tree depth (2, 5, 6, 10, 15, and 20) and the number of trees (50, 100, 200, and 500). This resulted in 400 Random Forest models and 20 Naïve Bayes models of optimal configurations. Table VI shows the setup of the four top-performing models, selected from the 420 evaluated models, where M1, M2, M3, and M4 correspond to the model configurations achieving the highest AUC, AUPRC, F-measure, and G-mean, respectively. We use the abbreviations RF, χ^2, and IG to represent Random Forest, Chi-Squared, and Information Gain, respectively.

TABLE VI
OPTIMAL MODEL CONFIGURATIONS BASED ON AUC, AUPRC, F-MEASURE, AND G-MEAN

Hyperparameter	**M1**	**M2**	**M3**	**M4**
Classifier	RF	RF	RF	RF
Feature Count	100	50	10	100
Feature Selection	χ^2	IG	IG	χ^2
RUS Ratio	1:1	1:1	1:1	1:1
Maximum Depth	10	5	6	10
Number of Trees	50	50	200	200

Table VII reports the performance of the top-performing models identified in Table VI. The results are highly similar across all four models, indicating that multiple configurations can achieve comparable predictive performance. Rather than choosing a single optimal model, these findings demonstrate the stability of model performance across closely related configurations and suggest that ML models can achieve a reliable level of predictive performance with tabular data alone. As such, these results provide a useful baseline for future studies, particularly those seeking to use additional data modalities to further improve prediction of cardiac mortality in patients with CHF. When using the MUSIC dataset, it is strongly suggested to use Random Forest, Information Gain or Chi-Squared as the feature selection technique, a feature count of 50 or 100, and a class ratio of 1:1. However, Table VII shows a lower number of features, such as 10, could achieve an optimal performance.

TABLE VII
PERFORMANCES OF OPTIMAL MODELS BASED ON AUC, AUPRC, F-MEASURE, AND G-MEAN

Performance Metric	**M1**	**M2**	**M3**	**M4**
AUC Score	**0.778**	0.774	0.755	0.775
AUPRC Score	0.772	**0.784**	0.734	0.780
F-Measure	0.698	0.673	**0.718**	0.713
G-Mean	0.698	0.670	0.711	**0.718**
False Positive Rate	0.304	0.339	0.312	**0.273**
False Negative Rate	0.296	0.312	**0.252**	0.287

C. Feature Importance

Finally, we examine the selected top 20 features under the best-performing configurations of Information Gain and Chi-Squared with a 1:1 RUS ratio. This analysis contextualizes the effects of the optimal configurations and highlights highly ranked predictors rather than providing a comprehensive feature importance study. According to Information Gain, laboratory variables were most represented, including Pro-BNP, hemoglobin, creatinine, albumin, protein, T3, and LDL, reflecting notable biomarkers of cardiac health. Echocardiographic measures such as left ventricular ejection fraction (LVEF), left atrial size, and left ventricular end-diastolic diameter were also present, demonstrating key structural and functional indicators. Several selected features were derived from a high-resolution or Holter ECG, including QRS duration, Q-waves, standard deviation of NN intervals (SDNN), bradycardia, and the number of supraventricular premature beats over 24 hours. Age was the primary demographic variable selected, and cardiothoracic ratio was the only radiographic feature in the top 20 ranking.

The Chi-Squared results place the number of ventricular premature contractions per hour as the top-ranked feature by a substantial margin. Other ECG-derived measurements include QRS duration, RR interval measures, standard deviations of average NN intervals over 5-minute segments (SDANN), SDNN, PR interval, and extrasystole couplets. Additional features include laboratory measures—gamma-glutamyl transpeptidase, creatinine, and urea—and behavioral variables, such as alcohol consumption and cigarettes smoked per year. Despite these differences, eight features overlap in the top 20 selected by Information Gain and Chi-Squared: Pro-BNP, Creatinine, the number of supraventricular premature beats in 24 hours, QRS duration, SDNN, LVEF, left atrial size, and age. Overall, these results suggest that physiological, structural, and electrophysiological characteristics were most informative under the optimal configurations.

V. Conclusion

Chronic heart failure affects a large number of people globally, often resulting in either sudden cardiac death or pump failure death. This paper addresses this issue by predicting the likelihood of cardiac mortality for patients with CHF using the tabular data derived from patient records and ECG measurements in the MUSIC database [7]. Five ML classifiers were evaluated across various configurations based on feature count, feature selection technique, RUS ratio, maximum tree depth, and number of trees. The optimal model configurations use Random Forest with 10, 50, or 100 features chosen using Information Gain or Chi-Squared feature selection and a perfectly balanced class ratio. These results highlight the importance of selecting a moderate number of informative features and balancing the dataset when predicting a cardiac mortality outcome. While maximum tree depth showed statistically significant differences, no single depth consistently achieved the highest-performing group across all four metrics, preventing a specific recommendation. Among the 4,704 models, Random Forest consistently demonstrated strong mean performance relative to the other classifiers, indicating that ensemble-based approaches may be particularly well-suited for this type of clinical prediction task. Finally, the optimal models achieve moderate predictive accuracy with AUC and AUPRC scores from 0.73 to 0.78. Future work will incorporate the signal data available in the MUSIC database for multimodal modeling and compare model performance to ML approaches on similar cardiac mortality datasets.

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